



Lab Sheet-01
Object Oriented Programming Using Java
MCA III Semester

1. Write a Java program to display your name, age, and college name using variables of appropriate data types.
2. Input two integers and print the result of all arithmetic operations (addition, subtraction, multiplication, division, and modulus).
3. Write a Java program to check whether a given number is even or odd using the modulus operator and if-else.
4. Write a program to demonstrate the use of relational operators by comparing two numbers.
5. Accept a floating-point number and convert it explicitly to an integer. Print both the original and the converted values.
6. Write a program to accept a character and print its ASCII value.
7. Input a number and check if it is divisible by both 3 and 5 using logical operators.
8. Input a character and check whether it is a vowel or a consonant using if-else.
9. Input marks of three subjects, calculate total and percentage, and print pass/fail using if-else (passing criteria: $\geq 40\%$ in all subjects).
10. Write a program to find the greatest of two numbers using if-else.
11. Input three numbers and print the largest using nested if-else.
12. Accept a year and check whether it is a leap year using if-else.
13. Accept a number and check whether it is positive, negative, or zero using if-else.
14. Accept a character and check if it is a digit, uppercase letter, lowercase letter, or a special character using character classification and if-else.
15. Accept a 3-digit number and check whether it is an Armstrong number using arithmetic operators and if-else.
16. Input salary and calculate tax based on income slabs using if-else:
 - $\text{Income} \leq 250000$: No tax
 - 250001 to 500000: 5%
 - 500001 to 1000000: 20%
 - Above 1000000: 30%
17. Accept an integer and check whether it is a palindrome using arithmetic operations and type casting.
18. Accept a number and print whether it is prime or not using if-else.



FORMERLY KNOWN AS UETR

19. Accept a character and use if-else to identify whether it's a valid operator (+, -, *, /) and perform the corresponding operation on two input numbers.
20. Accept age and gender and determine eligibility for marriage (male ≥ 21 , female ≥ 18) using nested if-else conditions.

